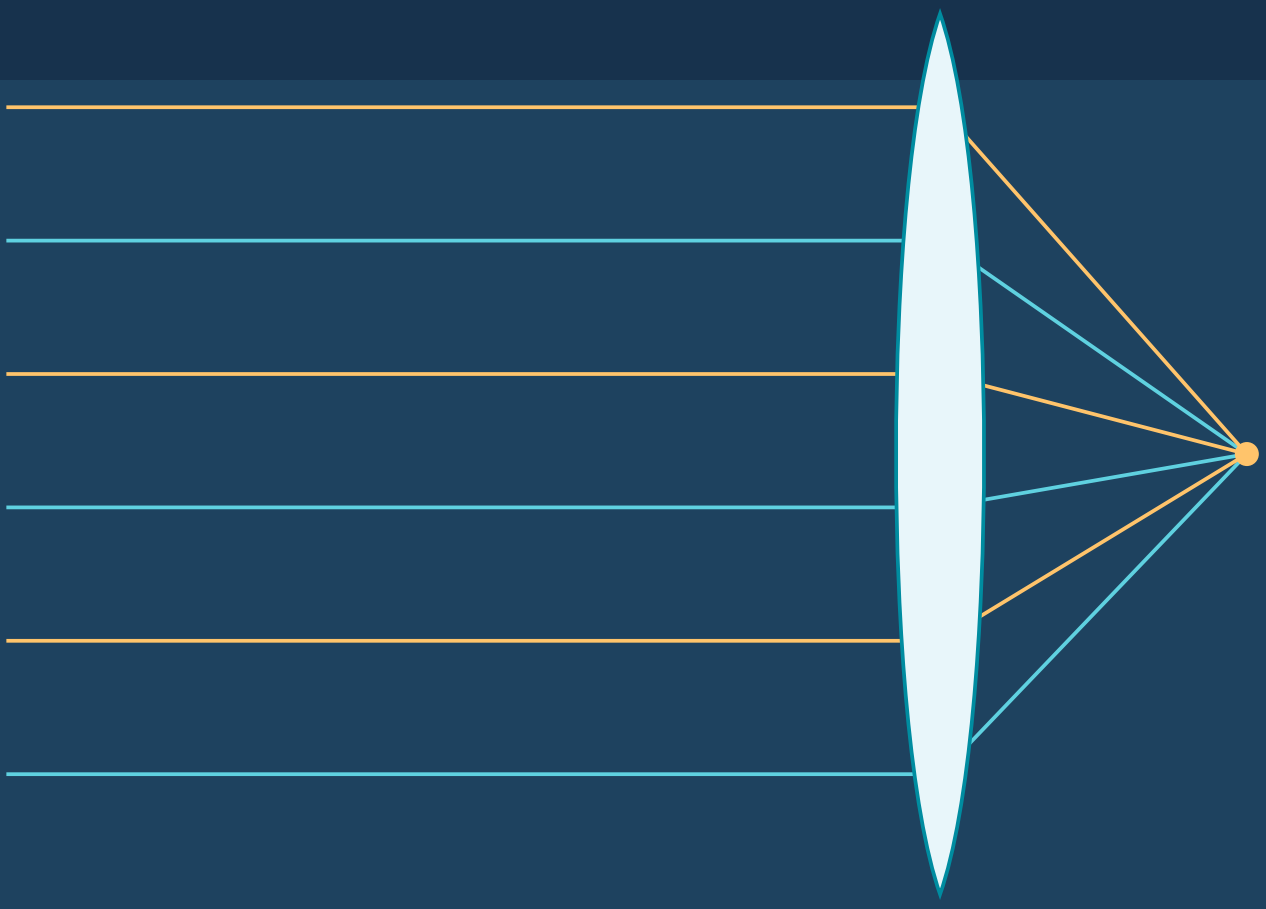
Light — Reflection & Refraction

Complete notes in simple English

CHAPTER 9 IN THE CURRENT NCERT BOOK · CHAPTER 10 IN SOME OLDER EDITIONS

Easy explanations | Ray diagrams | Formula sheet | Solved numericals | Practical work | Practice + answers | 10-minute revision



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1. Chapter roadmap and CBSE scope

This PDF gives **complete, original notes** for **Light — Reflection and Refraction**. It follows the NCERT Class 10 Science chapter and the CBSE Class X Science (086) scope. In the current 2026–27 NCERT book it is **Chapter 9**; many older books call the same chapter **Chapter 10**. Do not worry about the number — learn the ideas below.

The language is kept simple on purpose. Every important word is explained, formulas are broken into steps, and ray diagrams are drawn from the basic rules.

1.1 What you will be able to do

- Explain reflection by plane and curved surfaces.
- Draw and read ray diagrams for **concave and convex mirrors**.
- Use the **mirror formula** and **magnification** with correct signs.
- Explain refraction, refractive index, Snell's law and a glass slab.
- Draw and read ray diagrams for **convex and concave lenses**.
- Use the **lens formula**, magnification and **power of a lens**.
- Solve board-style numericals without mixing up signs or units.
- Answer practical/lab questions and avoid the most common mistakes.

1.2 Exactly what is in the syllabus

Table 1.1 — Chapter 9 scope in simple words

Part	NCERT/CBSE topics	What is expected
Mirrors	Reflection by curved surfaces; centre of curvature, principal axis, principal focus, focal length; images; mirror formula; magnification	Draw ray diagrams and solve numericals. **Derivation is not required.**
Refraction	Laws of refraction; refractive index; rectangular glass slab	Explain bending, use Snell's law and refractive index.
Lenses	Spherical lenses; image formation; lens formula; magnification; power of a lens	Draw ray diagrams and solve numericals. **Derivation is not required.**
Applications	Uses of spherical mirrors and lenses	Give a use and the reason why that mirror/lens is chosen.
Practical	Focal length of concave mirror and convex lens; ray tracing through a glass slab	Know the steps, observations, result and precautions.

Do not mix the next chapter here
The **human eye, defects of vision, prism, dispersion and scattering** are in the next chapter, The Human Eye and the Colourful World. They are not part of this chapter.

1.3 Best way to study this chapter

1. First learn the **words**: pole, focus, centre, normal, optical centre and principal axis.
2. Learn only **three or four ray rules**. Every diagram is made from those rules.
3. Practise the **New Cartesian sign convention** on a small axis. Most numerical mistakes are sign mistakes.
4. Keep one formula sheet beside you: mirror and lens formulas look similar but are not the same.
5. After every numerical, write one sentence about the image: real/virtual, erect/inverted, magnified/reduced.
6. Redraw each diagram once without looking. This takes less time than memorising six image tables.

2. Light and reflection: the foundation

2.1 A few basic ideas

- **Light** is a form of energy. We see an object when light from that object reaches our eyes.
- In one uniform medium, light travels in a straight line. This is called **rectilinear propagation**.
- A **ray** is a line with an arrow that shows the direction of light. A **beam** is a group of rays.
- A **transparent** material lets most light pass through; a **translucent** material lets some light through; an **opaque** material does not let light pass through.
- When light falls on a surface and returns to the same medium, the process is called **reflection**.

2.2 Reflection and its two laws

Look at a ray falling on a plane mirror. The ray that comes in is the **incident ray**. The ray that goes back is the **reflected ray**. The point where the ray strikes is the **point of incidence**. A line drawn perpendicular to the mirror at that point is the **normal**.

1. The **incident ray, reflected ray and normal** at the point of incidence all lie in the **same plane**.
2. The **angle of incidence (i)** is always equal to the **angle of reflection (r)**: `i = r`.

> **Very important**
> Both angles are measured from the **normal**, not from the mirror surface. If the ray falls along the normal, `i = 0°` and `r = 0°`, so it returns on the same path.

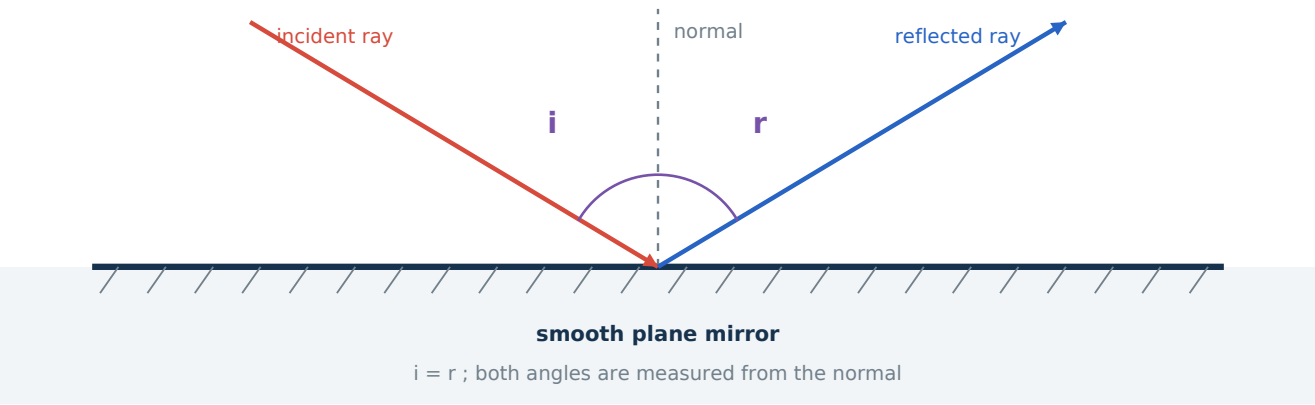


Figure 2.1 — Reflection at a plane surface. Angles i and r are measured from the dashed normal.

2.3 Regular and diffused reflection

Table 2.1 — Both kinds obey the same laws of reflection

Regular reflection	Diffused (irregular) reflection
Takes place on a smooth, polished surface.	Takes place on a rough or uneven surface.
Parallel rays remain parallel after reflection.	Parallel rays go in different directions after reflection.
A clear image can be formed.	No clear image is formed, but the surface becomes visible.
Example: plane mirror, still water.	Example: paper, wall, cloth, rough road.

Easy meaning: rough paper has many tiny surfaces with different normals. Each ray follows `i = r` at its own point, so the full beam spreads out. The law is not broken.

2.4 Real image, virtual image and magnification

Table 2.2 — Words used in every ray diagram

Term	Simple meaning	Quick test
Real image	Light rays actually meet after reflection/refraction.	Can be obtained on a screen.
Virtual image	Rays only appear to meet when produced backwards.	Cannot be obtained on a screen.
Erect image	Upright, in the same direction as the object.	The arrow points upward.
Inverted image	Upside down compared with the object.	The arrow points downward.
Magnification (m)	How many times the image height is compared with object height.	`m = image height / object height`.

For a **plane mirror**, the image is virtual, erect, the same size as the object, laterally inverted, and as far behind the mirror as the object is in front. Its magnification is `m = +1`. This is quick recall from earlier classes; the main exam work in this chapter is on curved mirrors and lenses.

3. Spherical mirrors: parts and meaning

3.1 What is a spherical mirror?

A **spherical mirror** is a mirror whose reflecting surface is a part of a hollow sphere. Think of a shiny ball cut into a small piece.

Table 3.1 — The two spherical mirrors

Concave mirror	Convex mirror
Reflecting surface is curved **inward**, like the inside of a spoon.	Reflecting surface bulges **outward**, like the back of a spoon.
Converges parallel rays after reflection.	Diverges parallel rays after reflection.
Focus is **real**: rays really meet in front of the mirror.	Focus is **virtual**: rays appear to come from behind the mirror.
Can form real or virtual images depending on object position.	Always forms a virtual, erect and diminished image for a real object.

3.2 Terms you must label correctly

- **Pole (P):** the centre point of the mirror surface. All distances are measured from P.
- **Centre of curvature (C):** the centre of the sphere of which the mirror is a part. It is not on the mirror.
- **Radius of curvature (R):** distance between P and C.
- **Principal axis:** the straight line joining P and C.
- **Aperture:** the width of the mirror's reflecting surface. Small-aperture diagrams are easier and more accurate.
- **Principal focus (F):** the point where rays parallel and close to the principal axis meet (concave) or appear to come from (convex).
- **Focal length (f):** distance between P and F.
- For a spherical mirror of small aperture: `R = 2f`, so `f = R / 2`.

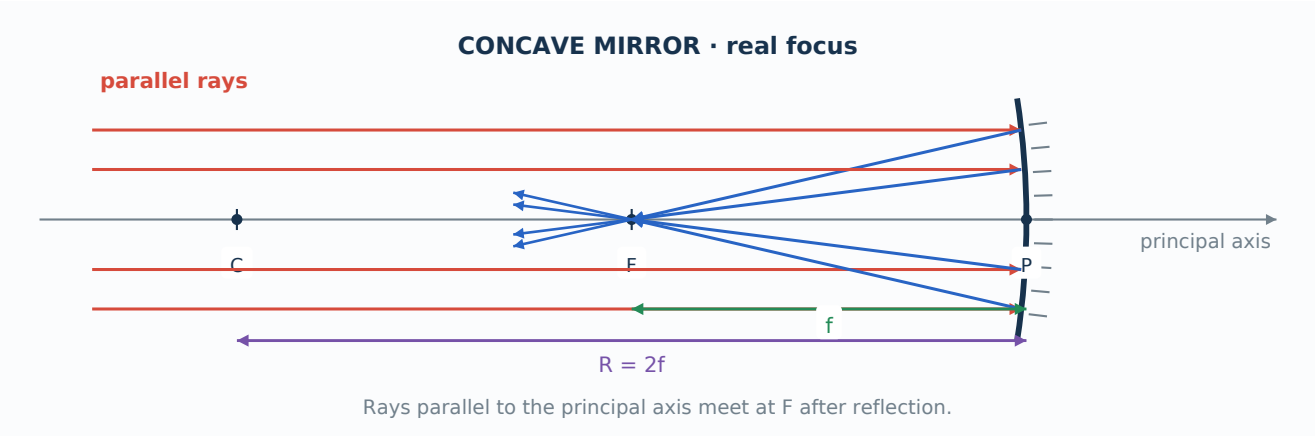


Figure 3.1 — A concave mirror brings parallel rays to its real focus F. C, F and P lie on the principal axis.

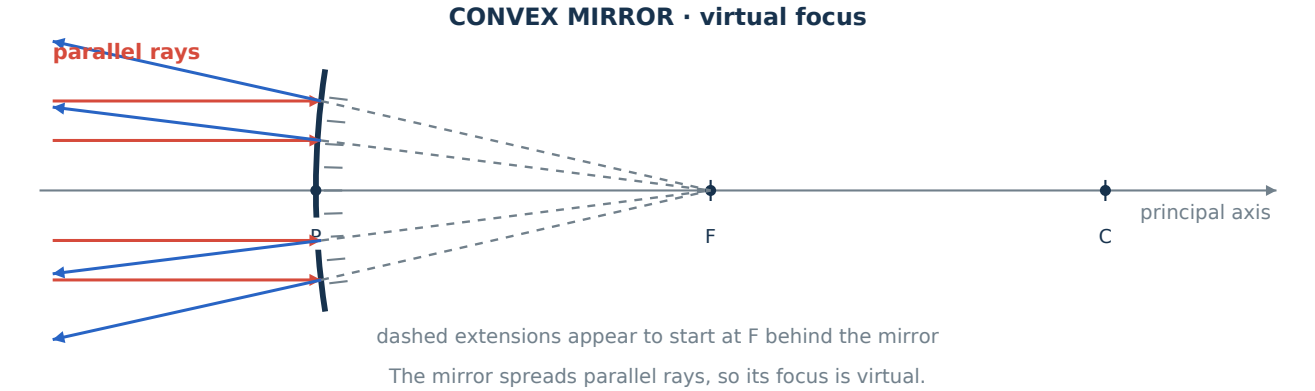


Figure 3.2 — A convex mirror spreads parallel rays. Their backward extensions appear to come from virtual focus F.

3.3 Why do the rays converge or diverge?

At every point of a curved mirror, draw a **normal** along the radius to C. The incident ray and reflected ray must make equal angles with that local normal. On a concave mirror the surface is shaped so these reflected rays come together. On a convex mirror they spread apart. This is simply the law of reflection applied many times.

3.4 Everyday uses — with the reason

Table 3.2 — Applications of spherical mirrors

Device	Mirror/lens	Why it is used
Torch, searchlight, car headlight	Concave mirror	A bulb at F gives a strong, nearly parallel beam after reflection.
Shaving mirror / makeup mirror	Concave mirror	Face between F and P gives a large, erect, virtual image.
Dentist's mirror	Concave mirror	Gives an enlarged erect image of a tooth when held close.
Solar furnace / solar cooker	Concave mirror	Concentrates sunlight at the focus and produces heat.
Vehicle rear-view mirror	Convex mirror	Always erect and gives a wide field of view because the image is small.
Shop/security mirror at corners	Convex mirror	Shows a large area in one small mirror.

4. Ray diagrams and images formed by mirrors

4.1 Four useful rays — learn these, not the full tables

To locate an image, draw any **two** of these rays from the top of the object. Where the reflected rays meet, or appear to meet, is the top of the image.

1. **Ray parallel to the principal axis:** after reflection, it passes through F in a concave mirror; in a convex mirror it appears to come from F behind the mirror.
2. **Ray passing through F (concave) or directed towards F (convex):** after reflection, it becomes parallel to the principal axis.
3. **Ray passing through C (concave) or directed towards C (convex):** it falls normally on the mirror and returns along the same path.
4. **Ray striking the pole P:** it reflects obeying `i = r` with the principal axis as the normal.

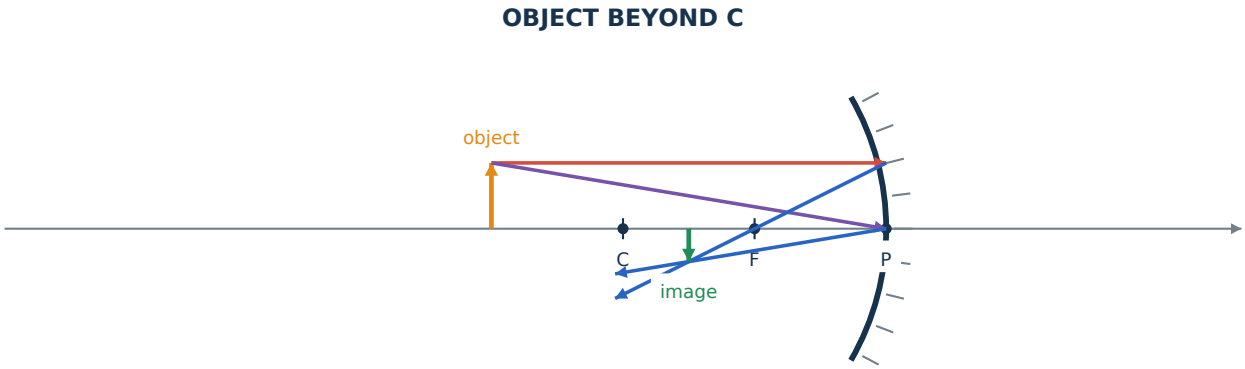
> **Drawing tip**
> Use a ruler, put arrows on every ray, and draw **dashed lines** for backward extensions behind a mirror. A virtual image is also usually drawn with dashed lines.

4.2 Concave mirror: complete image table

Table 4.1 — Images formed by a concave mirror

Object position	Image position	Nature and size
At infinity	At F	Real, inverted, highly diminished (point-sized).
Beyond C	Between F and C	Real, inverted, diminished.
At C	At C	Real, inverted, same size.
Between C and F	Beyond C	Real, inverted, enlarged.
At F	At infinity	Real, inverted, highly enlarged.
Between F and P	Behind the mirror	Virtual, erect, enlarged.

Memory line: as the object moves from infinity towards the concave mirror, the real image moves away from F and becomes larger. Once the object crosses F, the image suddenly becomes virtual and erect behind the mirror.



Real, inverted, diminished image between F and C

Figure 4.1 — Object beyond C: real, inverted and diminished image between F and C.

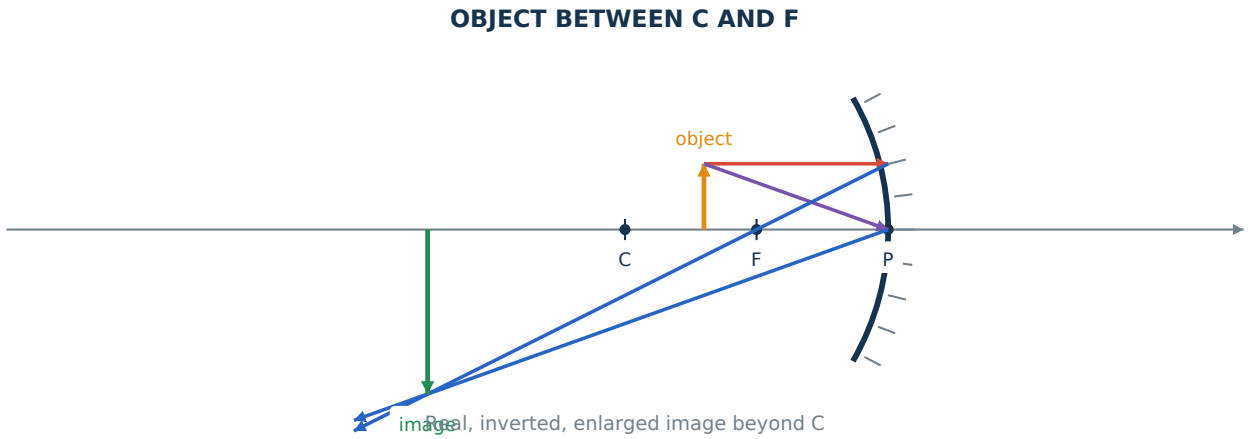


Figure 4.2 — Object between C and F: real, inverted and enlarged image beyond C.

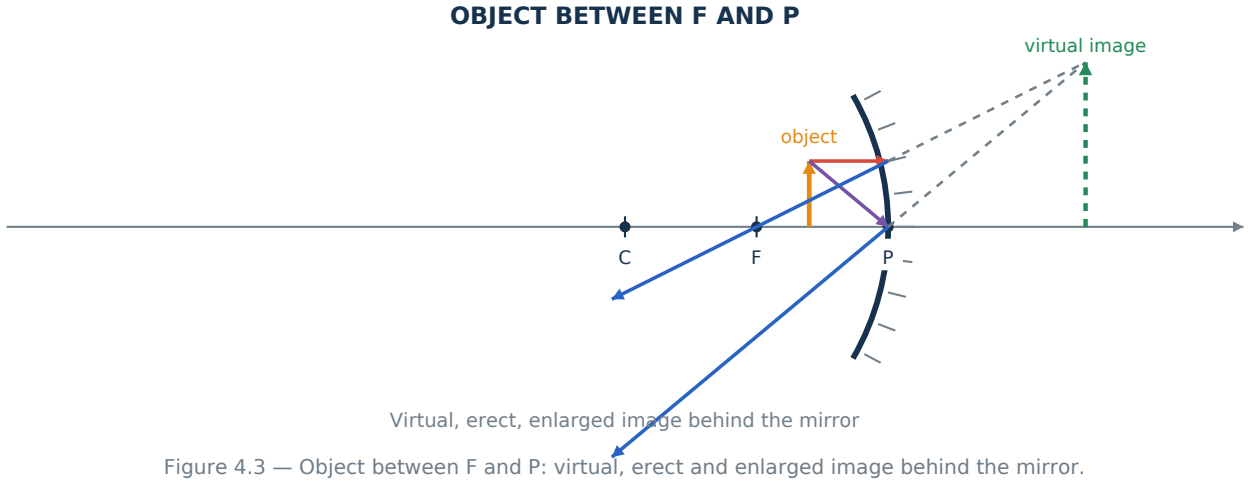


Figure 4.3 — Object between F and P: virtual, erect and enlarged image behind the mirror.

4.3 Convex mirror: complete image table

Table 4.2 — Images formed by a convex mirror

Object position	Image position	Nature and size
At infinity	At F, behind the mirror	Virtual, erect, highly diminished (point-sized).
Between infinity and P	Between P and F, behind the mirror	Virtual, erect, diminished.

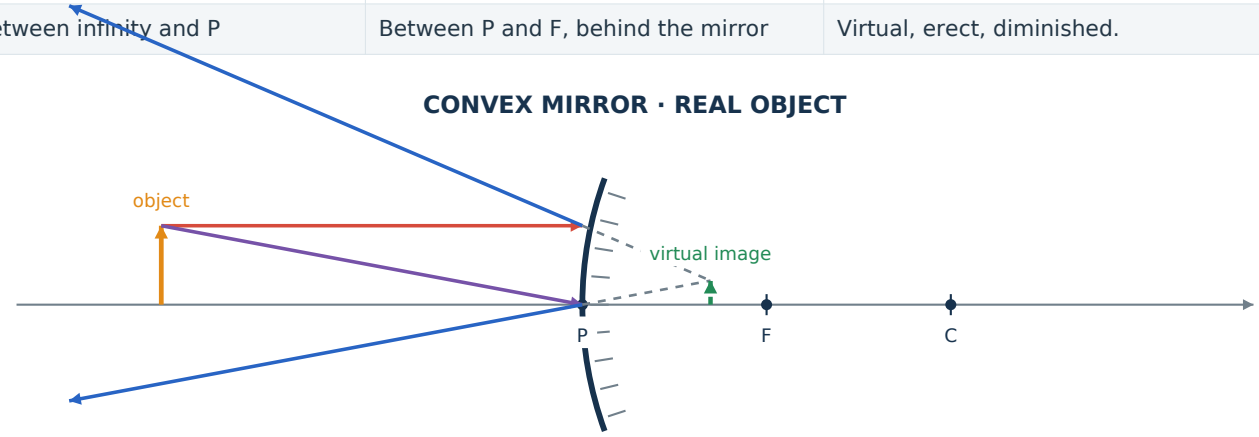


Figure 4.4 — A convex mirror always gives a virtual, erect and diminished image between P and F for a real object.

One-line memory

Concave: can give real/inverted or virtual/erect images. **Convex:** with a real object, always virtual, erect and smaller. This is why convex mirrors are used as rear-view mirrors.

5. Mirror formula, sign convention and magnification

5.1 New Cartesian sign convention for mirrors

Place the object on the **left** of the mirror so that incident light travels from left to right. Take the pole P as the origin and the principal axis as the x-axis.

- All distances are measured from **P**.
- Distances measured in the **direction of incident light** (to the right) are positive; distances to the left are negative.
- Heights measured **above** the principal axis are positive; heights below it are negative.
- The object is above the axis, so h is positive. A real inverted image has h' negative; a virtual erect image has h' positive.

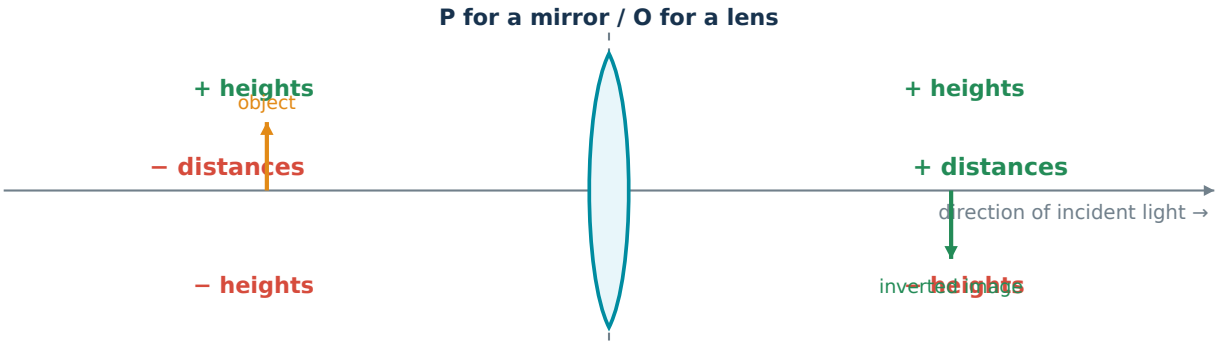


Figure 5.1 — The same sign rule is used for mirrors and lenses: right/above are positive, left/below are negative.

Table 5.1 — Signs for a mirror with the object on the left

Quantity	Common sign	Reason
Object distance u	Negative	Object is on the left of P.
Focal length f of concave mirror	Negative	Its focus is in front of the mirror, on the left.
Focal length f of convex mirror	Positive	Its focus is behind the mirror, on the right.
Image distance v, real image	Negative	Real image is in front, on the left.
Image distance v, virtual image	Positive	Virtual image is behind, on the right.
Image height h', inverted	Negative	Image lies below the principal axis.
Image height h', erect	Positive	Image lies above the principal axis.

5.2 Mirror formula and magnification

$$1/v + 1/u = 1/f$$

Mirror formula — use signs for u, v and f

$$m = h'/h = -v/u$$

Linear magnification produced by a spherical mirror

- u = object distance from P; v = image distance from P; f = focal length; h = object height; h' = image height.
- Use **one unit throughout** a problem: all cm or all m. Do not mix cm and m.
- m has no unit because it is a ratio of two lengths.
- |m| > 1 means enlarged; |m| < 1 means diminished; |m| = 1 means same size.
- For a real object in this chapter, m < 0 means real and inverted; m > 0 means virtual and erect.
- **Positive v for a mirror does not mean a real image.** For a mirror, a real image is on the same side as the object and has v < 0.

5.3 Worked mirror numericals

Example 5.1 — Concave mirror, real image

A 4 cm tall object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the image position, size and nature.

1. Given: u = -20 cm, f = -15 cm, h = +4 cm.
2. 1/v + 1/u = 1/f, so 1/v = 1/f - 1/u = -1/15 + 1/20 = -1/60.
3. Therefore v = -60 cm. The negative sign shows a real image 60 cm in front of the mirror.
4. m = -v/u = -(-60)/(-20) = -3 and h' = m × h = -3 × 4 = -12 cm.
5. **Answer:** real, inverted, 12 cm tall, 60 cm in front of the mirror.

Example 5.2 — Convex mirror, virtual image

An object is 20 cm in front of a convex mirror whose radius of curvature is 30 cm. Find the image position and magnification.

1. R = +30 cm, so f = R/2 = +15 cm; u = -20 cm.
2. 1/v = 1/f - 1/u = 1/15 + 1/20 = 7/60, so v = 60/7 ≈ +8.6 cm.
3. m = -v/u = -(60/7)/(-20) = 3/7 ≈ +0.43.
4. **Answer:** virtual and erect image about 8.6 cm behind the mirror, about 0.43 times the object size.

Example 5.3 — Object at C

A concave mirror has f = -20 cm and an object is placed at u = -40 cm. Find v and m.

1. 1/v = 1/f - 1/u = -1/20 + 1/40 = -1/40, so v = -40 cm.
2. m = -v/u = -(-40)/(-40) = -1.
3. **Answer:** image is at C, real, inverted and the same size. This matches the ray table.

Most common mirror mistakes
Using +f for a concave mirror, using the lens formula by accident, writing m = v/u instead of -v/u, or forgetting to convert R into f with f = R/2.

6. Refraction of light

6.1 What is refraction?

When light travels **obliquely** from one transparent medium to another, its speed changes and its path usually bends. This change in direction is called **refraction**.

Why does it bend? Light has different speeds in different materials. One side of a wavefront slows down or speeds up before the other side, so the whole ray turns. The colour/frequency stays the same; speed and wavelength change.

Table 6.1 — Three cases to remember

Change of medium	Speed	Direction of bending	Angle relationship
Rarer → denser (air → glass/water)	Decreases	Towards the normal	`r < i`
Denser → rarer (glass/water → air)	Increases	Away from the normal	`r > i`
Along the normal (`i = 0°`)	Changes	No change in direction	`r = 0°`

Optical density is not mass density
A medium with a larger refractive index is called **optically denser**. This does not mean it is physically heavier. For example, kerosene floats on water but is optically denser than water.

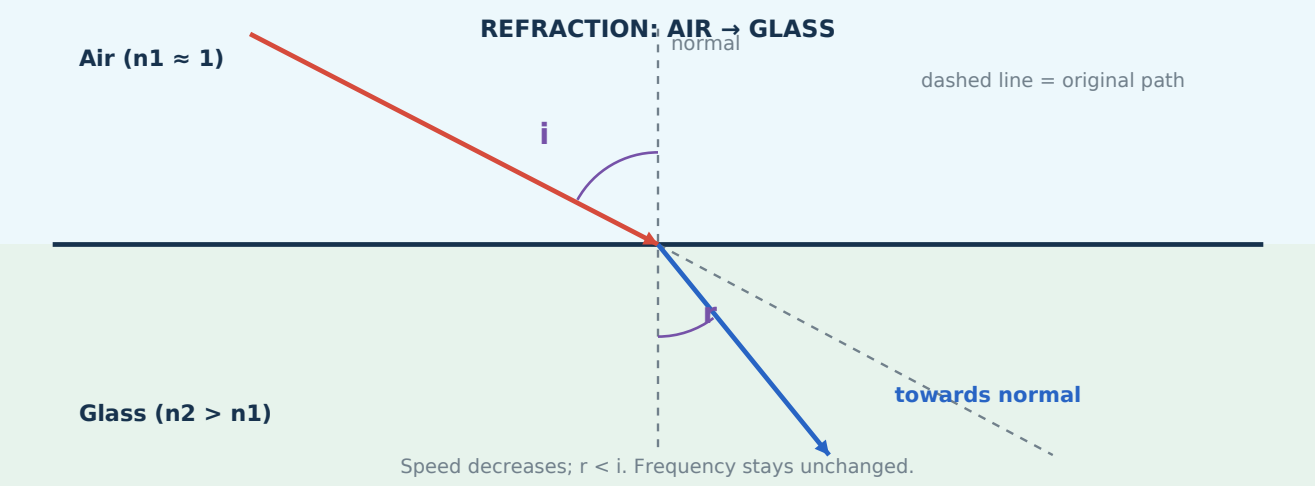


Figure 6.1 — Air to glass: light slows down and bends towards the normal. The dashed line shows the original straight path.

6.2 Laws of refraction

1. The **incident ray, refracted ray and normal** at the point of incidence all lie in the **same plane**.
2. For a given pair of media and a given colour of light, the ratio `sin i / sin r` is constant. This is **Snell's law**.

$$n_1 \sin i = n_2 \sin r$$

General form of Snell's law

$$n_{21} = \frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

Relative refractive index of medium 2 with respect to medium 1

Here `n1` and `n2` are absolute refractive indices, and `v1` and `v2` are the speeds of light in media 1 and 2. Also, `n12 = 1/n21`.

6.3 Absolute refractive index

The **absolute refractive index** of a medium tells us how much slower light travels in that medium than in vacuum (or, approximately, air).

$$n = c / v$$

Absolute refractive index; c = speed of light in vacuum, v = speed in the medium

- n has **no unit** because it is a ratio of two speeds.
- Vacuum has n = 1; air is about 1.0003 and is usually taken as 1.
- A larger n means a lower speed of light and a stronger bending effect.
- For ordinary transparent media, absolute n is greater than 1. A **relative** refractive index can be less than 1.

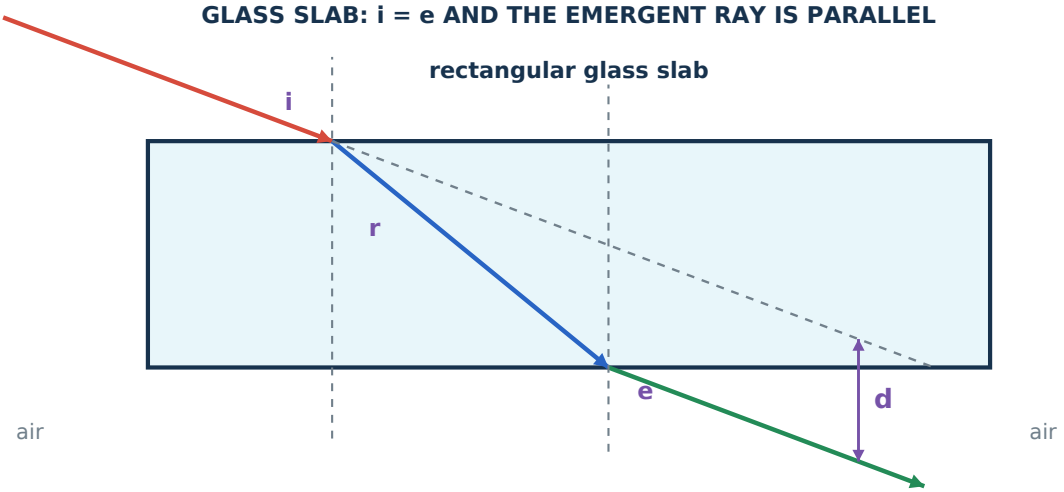
Table 6.2 — Common values (small variations are normal)

Medium	Approximate refractive index n	Easy comparison
Vacuum	1.00	Light has its maximum speed.
Air	1.0003 ≈ 1.00	Almost the same as vacuum.
Water	1.33	Light travels slower than in air.
Crown glass	1.52	Bends light more than water does.
Diamond	2.42	Light slows a lot; strong optical effects.

6.4 Refraction through a rectangular glass slab

A rectangular glass slab has two **parallel** faces. Light refracts once when it enters the slab and again when it leaves. The second bending cancels the change in direction from the first bending.

- At air → glass, the ray bends **towards** the normal.
- At glass → air, it bends **away** from the normal.
- The **emergent ray is parallel** to the incident ray: i = e.
- The emergent ray is shifted sideways. This perpendicular shift is called **lateral displacement**.
- A thicker slab, a larger angle of incidence or a larger refractive index generally gives a larger shift.



d = lateral displacement (sideways shift)

Figure 6.2 — Refraction through a rectangular glass slab: the emergent ray is parallel to the incident ray but laterally displaced.

6.5 Worked refraction numericals

Example 6.1 — Speed of light in glass

1. Given `n_glass = 1.50` and `c = 3 × 10^8 m/s`.
2. `n = c/v`, so `v = c/n = (3 × 10^8)/1.50 = 2 × 10^8 m/s`.
3. **Answer:** light travels at `2 × 10^8 m/s` in the glass.

Example 6.2 — Finding refractive index from angles

Light goes from air into a transparent material. The angle of incidence is `30°` and the angle of refraction is `19°`. Estimate the refractive index of the material.

1. `sin 30° = 0.500` and `sin 19° ≈ 0.326`.
2. `n ≈ sin i / sin r = 0.500/0.326 ≈ 1.53`.
3. **Answer:** about `1.53`, close to crown glass.

Example 6.3 — Relative refractive index

1. Given `n_glass = 1.52` and `n_water = 1.33`.
2. Glass with respect to water: `n_gw = n_glass/n_water = 1.52/1.33 ≈ 1.14`.
3. Water with respect to glass: `n_wg = 1/n_gw ≈ 0.88`.

7. Spherical lenses: parts and ray rules

7.1 Convex and concave lenses

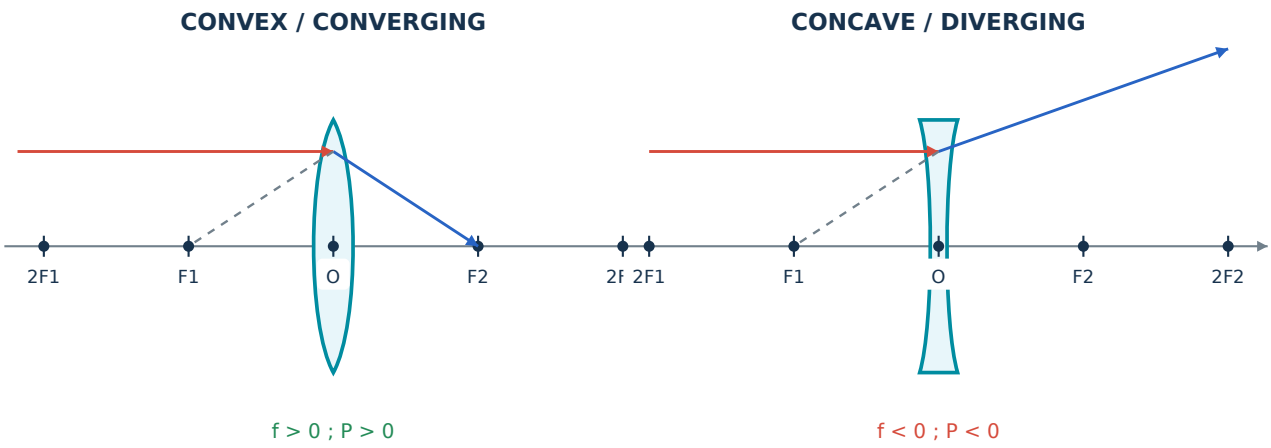
A **lens** is a piece of transparent material bounded by two spherical surfaces. It works by refraction, not reflection.

Table 7.1 — The two spherical lenses

Convex lens	Concave lens
Thicker in the middle and thinner at the edges.	Thinner in the middle and thicker at the edges.
Converging lens: brings parallel rays together.	**Diverging lens**: spreads parallel rays apart.
Has a real principal focus on the other side of the lens.	Has a virtual principal focus on the object side.
Can form real/inverted or virtual/erect images.	Always forms a virtual, erect and diminished image for a real object.
`f` is positive; power is positive.	`f` is negative; power is negative.

7.2 Points and distances on a lens diagram

- **Optical centre (O):** the central point of a thin lens. A ray passing through O goes nearly undeviated.
- **Principal axis:** the straight line passing through O and the two centres of curvature.
- **Aperture:** the effective diameter of the lens.
- **First focus F1:** focus on the object side. **Second focus F2:** focus on the other side.
- **Focal length f:** distance from O to F1 or F2. For a thin lens in air, both sides have the same magnitude of f.
- **2F1 and 2F2:** points at a distance `2f` from O on the two sides. They are very useful in ray diagrams.

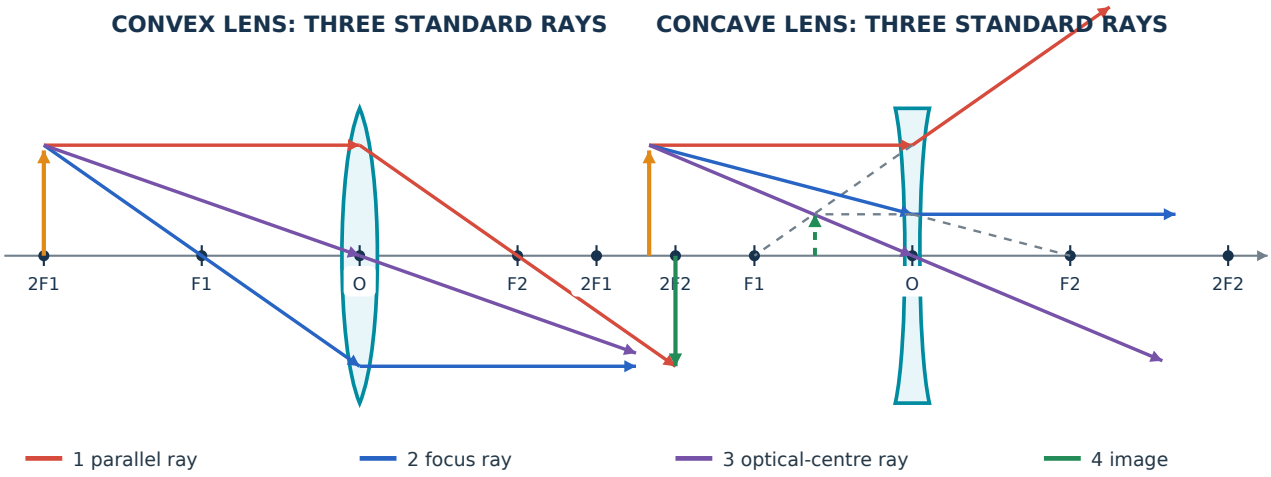


F1/F2 are focal points; 2F1/2F2 are at twice the focal distance from O.

Figure 7.1 — Convex and concave lenses with O, F1, F2, 2F1 and 2F2 marked on the principal axis.

7.3 Three ray rules for lenses

1. A ray **parallel to the principal axis** passes through F2 after a convex lens; after a concave lens it appears to come from F1.
2. A ray passing through F1 before a convex lens emerges **parallel** to the axis. A ray directed towards F2 before a concave lens emerges parallel.
3. A ray passing through the **optical centre O** continues almost without deviation for a thin lens.



Dashed lines are backward extensions; use any two rays to locate an image.

Figure 7.2 — Three standard rays used to construct lens ray diagrams. Dashed lines are backward extensions.

7.4 Everyday uses

Table 7.2 — Applications of spherical lenses

Use	Lens	Reason
Magnifying glass	Convex	Object inside f gives a virtual, erect and enlarged image.
Camera and projector	Convex	Can form a real image on film/sensor or a screen.
Simple microscope/telescope	Convex	Combines enlarged real/virtual images.
Spectacle lens for far-sightedness	Convex	Converges incoming rays; details are in the next chapter.
Spectacle lens for near-sightedness	Concave	Diverges incoming rays; details are in the next chapter.
Door peephole / some wide-view systems	Concave	Gives a small erect image of a wider area.

8. Ray diagrams and images formed by lenses

8.1 Convex lens: complete image table

Table 8.1 — Images formed by a convex lens

Object position	Image position	Nature and size
At infinity	At F2	Real, inverted, highly diminished (point-sized).
Beyond 2F1	Between F2 and 2F2	Real, inverted, diminished.
At 2F1	At 2F2	Real, inverted, same size.
Between F1 and 2F1	Beyond 2F2	Real, inverted, enlarged.
At F1	At infinity	Real, inverted, highly enlarged.
Between F1 and O	On the object side	Virtual, erect, enlarged.

Memory line: for a convex lens, a real inverted image becomes larger as the object moves from far away towards F1. When the object crosses F1, the image becomes virtual, erect and enlarged on the object side.

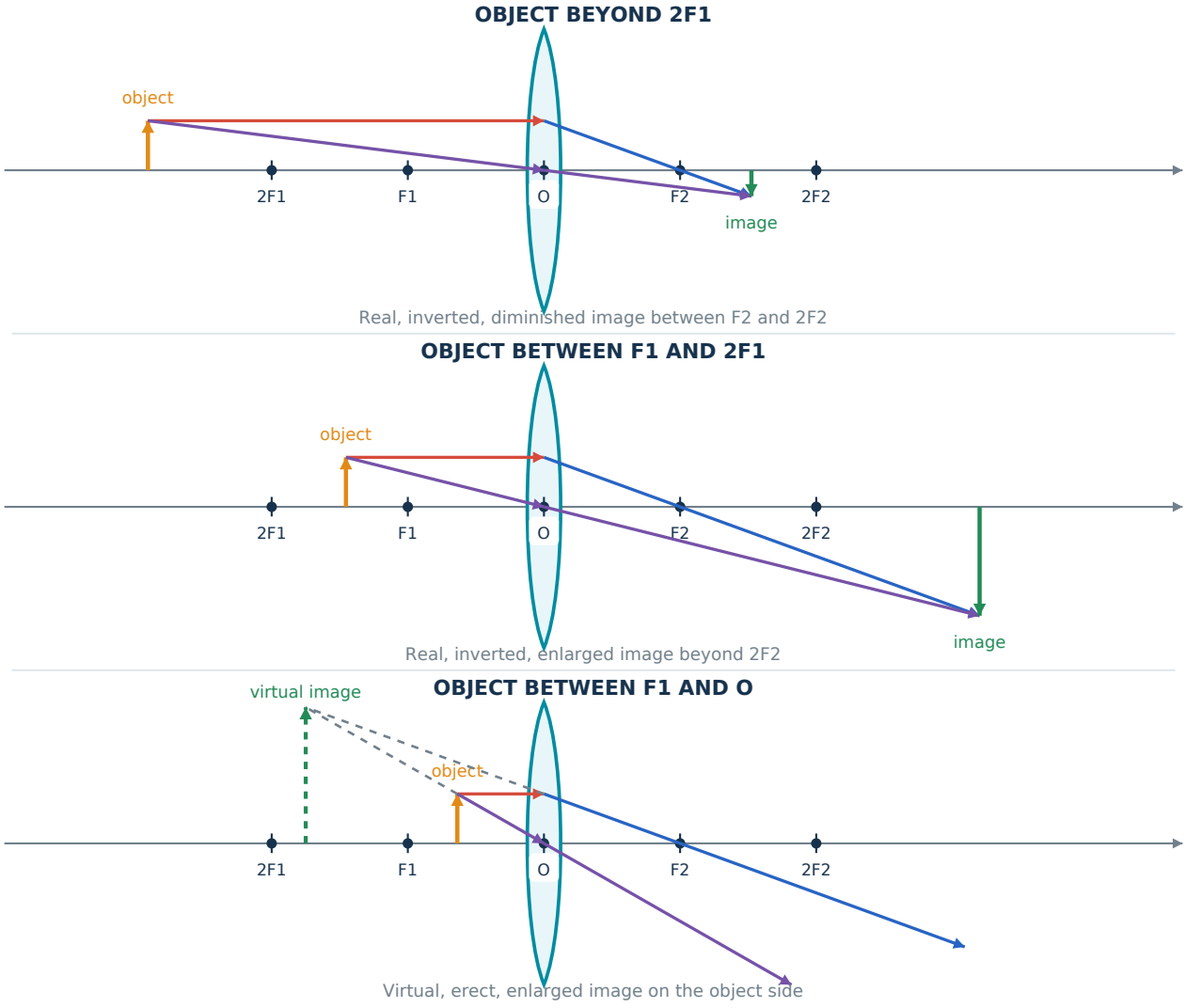


Figure 8.1 — Three important convex-lens cases: object beyond 2F1, between F1 and 2F1, and inside F1.

8.2 Concave lens: complete image table

Table 8.2 — Images formed by a concave lens

Object position	Image position	Nature and size
At infinity	At F1	Virtual, erect, highly diminished (point-sized).
Between infinity and O	Between O and F1	Virtual, erect, diminished.

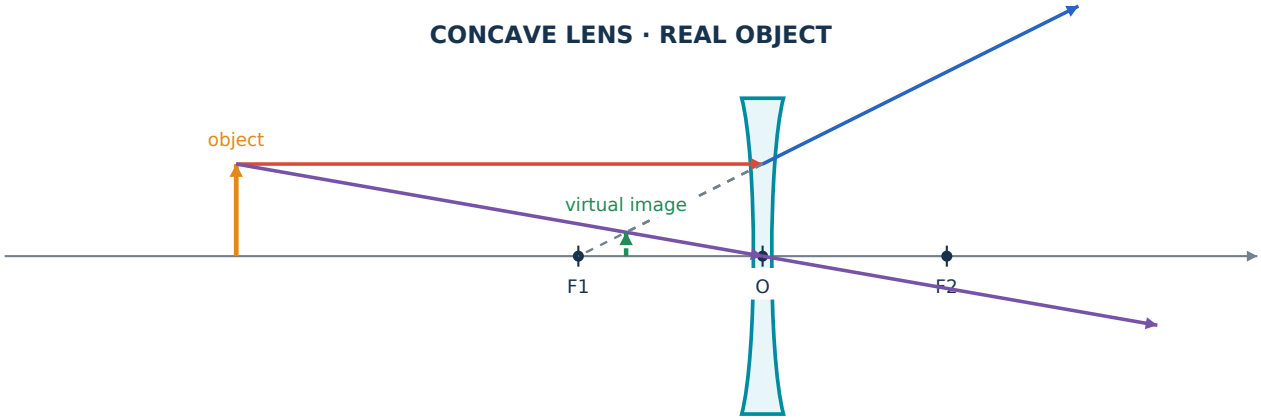


Image is always virtual, erect and smaller, between O and F1.

Figure 8.2 — A concave lens forms a virtual, erect and diminished image between O and F1.

Compare the two diverging devices
A **convex mirror** and a **concave lens** both always give virtual, erect, diminished images for a real object. A concave mirror and convex lens can give both real and virtual images.

9. Lens formula, magnification and power

9.1 Sign convention for lenses

Keep the object on the **left** and let incident light travel left to right. Take O as the origin. Distances to the right are positive and distances to the left are negative; heights above the axis are positive and below are negative.

Table 9.1 — Signs for a lens with the object on the left

Quantity	Common sign	Reason
Object distance u	Negative	Object is on the left of O.
Focal length f of convex lens	Positive	Second focus F2 is on the right.
Focal length f of concave lens	Negative	Its focus is virtual, on the left.
Image distance v, real image	Positive	Real image is formed on the right.
Image distance v, virtual image	Negative	Virtual image is on the left.

9.2 Lens formula and magnification

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Lens formula — notice the minus sign

$$m = \frac{h'}{h} = \frac{v}{u}$$

Linear magnification produced by a spherical lens

- Use one length unit throughout: all cm or all m.
- `m < 0` means inverted and real; `m > 0` means erect and virtual (for a real object).
- `|m| > 1`, `|m| < 1` and `|m| = 1` mean enlarged, diminished and same size respectively.
- The mirror magnification has `-v/u`; the lens magnification has `v/u`. Do not interchange them.

9.3 Power of a lens

The **power** of a lens tells us how strongly it converges or diverges light. A lens with a short focal length bends light more and has greater power.

P = 1 / f (in metres)

Power in dioptres

P = 100 / f (in cm) and P_total = P1 + P2 + ...

Quick conversion and thin lenses in contact

- The SI unit of power is the **dioptre (D)**, where `1 D = 1 m^-1`.
- A convex lens has **positive** power; a concave lens has **negative** power.
- **One dioptre** is the power of a lens of focal length `1 m`.
- For thin lenses kept in contact, simply add their powers with signs.
- If focal length doubles, power becomes half; if focal length halves, power doubles.

Table 9.2 — Power calculations

Focal length	Power	Type
`+0.50 m` or `+50 cm`	`+2 D`	Convex/converging
`+0.25 m` or `+25 cm`	`+4 D`	Convex/converging
`-0.40 m` or `-40 cm`	`-2.5 D`	Concave/diverging
`-2.00 m` or `-200 cm`	`-0.5 D`	Concave/diverging

9.4 Worked lens numericals

Example 9.1 — Convex lens, real image

A 5 cm object is 25 cm from a convex lens of focal length 10 cm. Find the image.

1. Given `u = -25 cm`, `f = +10 cm`, `h = +5 cm`.
2. `1/v - 1/u = 1/f`, so `1/v = 1/f + 1/u = 1/10 - 1/25 = 3/50`.
3. `v = +50/3 ≈ +16.7 cm`: a real image on the other side.
4. `m = v/u = (50/3)/(-25) = -2/3`; `h' = m × h ≈ -3.33 cm`.
5. **Answer:** real, inverted, about 3.33 cm tall, 16.7 cm from the lens.

Example 9.2 — Concave lens, virtual image

A 6 cm object is 30 cm from a concave lens of focal length 20 cm. Find the image.

1. Given `u = -30 cm`, `f = -20 cm`, `h = +6 cm`.
2. `1/v = 1/f + 1/u = -1/20 - 1/30 = -1/12`, so `v = -12 cm`.
3. `m = v/u = (-12)/(-30) = +0.4`; `h' = +2.4 cm`.
4. **Answer:** virtual, erect, 2.4 cm tall, 12 cm on the object side.

Example 9.3 — Lens power and focal length

1. A lens has `P = +4 D`. Then `f = 1/P = +0.25 m = +25 cm`; it is convex.
2. A lens has `f = -50 cm = -0.50 m`. Then `P = 1/f = -2 D`; it is concave.

Example 9.4 — Two lenses in contact

Thin lenses of powers `+2.5 D` and `-1.5 D` are placed in contact. Find the total power and focal length.

1. `P_total = +2.5 + (-1.5) = +1.0 D`.
2. `f_total = 1/P_total = +1.0 m = +100 cm`.
3. **Answer:** the combination behaves like a converging convex lens of focal length 1 m.

9.5 Mirror or lens? Keep the formulas separate

Table 9.3 — Formula comparison

Feature	Spherical mirror	Spherical lens
Main process	Reflection	Refraction
Formula	`1/v + 1/u = 1/f`	`1/v - 1/u = 1/f`
Magnification	`m = -v/u`	`m = v/u`
Concave device	Concave mirror: `f < 0`	Concave lens: `f < 0`
Convex device	Convex mirror: `f > 0`	Convex lens: `f > 0`
Power	Not asked in this chapter	`P = 1/f (m)`

10. Practical work and lab skills

10.1 Finding the focal length of a concave mirror

A very distant object sends almost parallel rays. A concave mirror brings those rays to its principal focus, so the distance from P to the sharp image gives approximately `f`.

1. Fix the concave mirror on a stand and point it at a distant tree or building. **Never use the Sun directly.**
2. Place a white screen in front of the reflecting surface, on the same side as the distant object.
3. Move the screen until a small, sharp, inverted image is formed.
4. Measure the distance from the pole P of the mirror to the screen. Repeat and take the mean.
5. **Result:** mean distance ≈ focal length `f` of the concave mirror.

10.2 Finding the focal length of a convex lens

1. Hold the convex lens facing a distant tree or building.
2. Put a white screen on the other side of the lens and move it until a sharp, small, inverted image appears.
3. Measure the distance from the optical centre O to the screen. Repeat and take the mean.
4. **Result:** mean distance ≈ focal length `f` of the convex lens.

10.3 Tracing a ray through a rectangular glass slab

1. Place the glass slab on white paper and draw its boundary.
2. Fix two pins in a straight line to represent the incident ray striking one face obliquely.
3. Looking from the opposite face, fix two more pins so that all four pins appear in one straight line.
4. Remove the slab and pins; join the pin positions to draw the incident and emergent rays.
5. Join the points of incidence and emergence to draw the refracted ray inside the slab. Draw normals and measure `i`, `r` and `e` with a protractor.

6. Repeat for another angle of incidence.

- **Observation:** i is approximately equal to e; the emergent ray is parallel to the incident ray.
- **Result:** the ray is laterally displaced because it bends towards the normal on entry and away from it on exit.
- **Precautions:** use a sharp pencil, keep pins vertical and well separated, use an oblique angle, and do not shift the slab while tracing.

Practical viva tip
If asked why a distant object is used, say: **rays from a distant object are nearly parallel, so they meet at the principal focus.** This one sentence answers both focal-length experiments.

11. Formula sheet and rapid revision

11.1 Every formula in one place

Table 11.1 — One-page formula sheet

Formula	Meaning	Important sign/unit point
$R = 2f$ or $f = R/2$	Radius and focal length of a spherical mirror	Use magnitude; apply sign separately.
$1/v + 1/u = 1/f$	Mirror formula	The sign between terms is **plus**.
$m = h'/h = -v/u$	Mirror magnification	The minus sign is part of the formula.
$n = c/v$	Absolute refractive index	No unit; ordinary transparent media have $n > 1$.
$n1 \sin i = n2 \sin r$	Snell's law	Angles are measured from the normal.
$n21 = n2/n1 = v1/v2 = \sin i/\sin r$	Relative refractive index	$n12 = 1/n21$.
$i = e$	Rectangular glass slab	Incident and emergent rays are parallel.
$1/v - 1/u = 1/f$	Lens formula	The sign between terms is **minus**.
$m = h'/h = v/u$	Lens magnification	There is **no minus sign** here.
$P = 1/f$ (m)	Power of a lens	f must be in metres; unit is dioptre (D).
$P = 100/f$ (cm)	Power when f is in centimetres	Keep the sign of f.
$P_{total} = P1 + P2 + ...$	Thin lenses in contact	Add powers with signs.

11.2 Last-day revision in simple lines

Mirrors

- Concave converges; convex diverges.
- Concave mirror can make real/inverted or virtual/erect images; convex mirror always gives virtual, erect, smaller images for a real object.
- Object at C gives image at C, same size and inverted.
- Object between F and P of a concave mirror gives a large erect virtual image.
- Mirror formula is $1/v + 1/u = 1/f$; mirror magnification is $-v/u$.

Refraction

- Air → glass/water: slower, towards normal. Glass/water → air: faster, away from normal.
- Normal incidence: speed changes but direction does not.
- $n = c/v$; larger n means lower speed.

- In a rectangular glass slab, the emergent ray is parallel to the incident ray and laterally shifted.

Lenses

- Convex converges and has `f > 0`, `P > 0`; concave diverges and has `f < 0`, `P < 0`.
- Convex lens can make real/inverted or virtual/erect images; concave lens always gives virtual, erect, smaller images for a real object.
- Object at 2F1 of a convex lens gives image at 2F2, same size and inverted.
- Lens formula is `1/v - 1/u = 1/f`; lens magnification is `v/u`.
- `P = 1/f (m)`; add powers for thin lenses in contact.

11.3 Tiny glossary

Table 11.2 — Words examiners expect you to know

Word	Easy meaning
Normal	Line at 90° to a surface at the point where light strikes.
Pole P	Middle point of a mirror surface.
Optical centre O	Middle point of a thin lens; a ray through it goes nearly straight.
Focus F	Point where parallel rays meet or appear to come from.
Principal axis	Main straight line through P/O and the focus.
Real image	An image that can be caught on a screen.
Virtual image	An image that cannot be caught on a screen.
Lateral displacement	Sideways shift of the emergent ray from a glass slab.
Dioptre	Unit of lens power; `1 D = 1 m^-1`.
Aperture	Useful width/diameter of a mirror or lens.

The one sign check that saves most marks
Object is on the left, so `u` is negative. Concave mirror and concave lens have negative `f`; convex mirror and convex lens have positive `f`. Then choose the correct mirror/lens formula.

12. How to solve numericals without fear

12.1 A five-step method

1. **Write the given values with signs and units.** Example: `u = -25 cm`, `f = -15 cm`.
2. **Choose the device:** mirror or lens. Write the correct formula before substituting.
3. **Substitute carefully.** Keep all lengths in cm or all in metres.
4. **Solve for the unknown.** Keep fractions until the last step to avoid rounding errors.
5. **Interpret the answer:** positive/negative `v`, `m` and `h'` tell you where the image is and what it looks like.

12.2 Worked mixed examples

Example 12.1 — Object inside the focus of a concave mirror

A concave mirror has radius 30 cm. An object is 10 cm in front of it. Find `v` and `m`.

1. `R = 30 cm`, so `f = -15 cm`; `u = -10 cm`.
2. `1/v = 1/f - 1/u = -1/15 + 1/10 = 1/30`, so `v = +30 cm`.
3. `m = -v/u = -30/(-10) = +3`.
4. **Answer:** virtual, erect and three times enlarged image 30 cm behind the mirror.

Example 12.2 — Convex lens with a distant object

A convex lens has `f = +20 cm`. An object is 60 cm away. Find the image position and magnification.

1. `1/v = 1/f + 1/u = 1/20 - 1/60 = 1/30`, so `v = +30 cm`.
2. `m = v/u = 30/(-60) = -0.5`.
3. **Answer:** real, inverted, half-sized image 30 cm on the other side of the lens.

Example 12.3 — Start from lens power

A lens has power `-2.5 D`. An 8 cm object is 60 cm from it. Find the image height.

1. `f = 1/P = -0.40 m = -40 cm`; the lens is concave.
2. `1/v = 1/f + 1/u = -1/40 - 1/60 = -1/24`, so `v = -24 cm`.
3. `m = v/u = (-24)/(-60) = +0.4`; `h' = 0.4 × 8 = +3.2 cm`.
4. **Answer:** virtual, erect, 3.2 cm tall.

Example 12.4 — Refractive index and speed

Light travels in a medium at `1.875 × 10^8 m/s`. Find its refractive index.

1. `n = c/v = (3 × 10^8)/(1.875 × 10^8) = 1.6`.
2. **Answer:** `n = 1.6`; light travels 1.6 times slower than in vacuum.

Five fast error checks
Did I use the **mirror** formula on a mirror question? Did I convert `R` to `f`? Did I put a **negative sign** on `u`? Did I use **metres** for power? Did I state the image nature at the end?

13. Practice questions (original, NCERT-style)

Try these without looking at the answer key. They are written in simple language but test the same ideas as board questions.

13.1 Multiple-choice questions

1. Which mirror can form a real and enlarged image? (a) Plane mirror (b) Concave mirror (c) Convex mirror (d) None of these.
2. A ray strikes a mirror along the normal. The angle of incidence is: (a) 90° (b) 0° (c) 45° (d) 180°.
3. For a convex mirror, the focal length is usually: (a) positive (b) negative (c) zero (d) equal to R.
4. The SI unit of power of a lens is: (a) metre (b) dioptre (c) watt (d) joule.
5. Light enters glass from air at an oblique angle. It bends: (a) towards the normal (b) away from the normal (c) back along the same path (d) not at all.
6. The correct lens formula is: (a) `1/v + 1/u = 1/f` (b) `1/v - 1/u = 1/f` (c) `v/u = -1/f` (d) `u - v = f`.
7. For a real object, a concave lens always forms: (a) a real enlarged image (b) a real smaller image (c) a virtual erect smaller image (d) a virtual inverted image.
8. For a spherical mirror, `m = -2` means the image is: (a) virtual and erect (b) real, inverted and enlarged (c) real, inverted and smaller (d) same size.
9. If `n = c/v`, a larger refractive index means: (a) larger speed (b) smaller speed (c) no change in speed (d) zero speed.
10. A lens has power `+4 D`. Its focal length is: (a) `+4 m` (b) `+0.25 m` (c) `-0.25 m` (d) `+25 m`.
11. The emergent ray from a rectangular glass slab is: (a) perpendicular to the incident ray (b) parallel to the incident ray (c) along the normal (d) always along the same straight line.
12. Which combination behaves as a converging lens? (a) `-2 D` and `-1 D` (b) `+3 D` and `-4 D` (c) `+2 D` and `+1 D` (d) `+1 D` and `-2 D`.

13.2 Assertion–Reason questions

Choose **(A)** if both statements are true and the Reason correctly explains the Assertion; **(B)** if both are true but the Reason is not the correct explanation; **(C)** if Assertion is true but Reason is false; **(D)** if Assertion is false but Reason is true.

1. **Assertion:** Convex mirrors are used as rear-view mirrors. **Reason:** They give erect images and a wider field of view.
2. **Assertion:** Light bends towards the normal when it enters glass from air. **Reason:** The speed of light becomes smaller in glass.
3. **Assertion:** A concave lens has negative power. **Reason:** Its focal length is negative under the usual sign convention.
4. **Assertion:** The magnification of a real image formed by a spherical mirror is positive. **Reason:** A real image formed by a spherical mirror is inverted.
5. **Assertion:** The incident and emergent rays of a rectangular glass slab are parallel. **Reason:** The two refracting faces are parallel and the medium on both sides is the same.
6. **Assertion:** Diamond has a very high refractive index. **Reason:** A ray through the optical centre of a thin lens is almost undeviated.

13.3 Short-answer questions

1. State the two laws of reflection and say from which line the angles are measured.
2. Give two reasons why a convex mirror is preferred as a vehicle rear-view mirror.
3. Define principal focus and focal length of a concave mirror. Write the relation between `R` and `f`.
4. State Snell's law and define absolute refractive index.
5. Why does the emergent ray from a rectangular glass slab remain parallel to the incident ray?
6. Define one dioptre. Why is the power of a concave lens negative?
7. Write two differences between a real image and a virtual image.
8. Why is a concave mirror used in a car headlight? Explain with a ray rule.
9. A student says, 'No bending means the speed did not change.' Is this always correct? Explain in one or two lines.

13.4 Numerical questions

1. A 4 cm object is 25 cm from a concave mirror of focal length 15 cm. Find `v`, `m`, `h'` and the nature of the image.
2. A convex mirror has `R = 40 cm`. An object is 30 cm in front of it. Find `v` and `m`.
3. The refractive index of glass is 1.50. Find the speed of light in the glass.
4. Light enters a medium from air with `i = 60°` and `r = 35°`. Estimate the refractive index. Use `sin 60° = 0.866` and `sin 35° = 0.574`.
5. A 3 cm object is 20 cm from a convex lens of focal length 15 cm. Find `v`, `m`, `h'` and the nature of the image.
6. A concave lens has `f = -15 cm`. An object is 45 cm away. Find `v` and `m`.
7. Two thin lenses of powers `+5 D` and `-2 D` are in contact. Find the total power and focal length.
8. Find the power of a lens whose focal length is `-80 cm`. State the type of lens.

13.5 Diagram/case-based questions

1. A concave mirror forms a virtual, erect and enlarged image of a candle. (a) Between which two points is the candle? (b) Name two rays you would draw to locate the image. (c) Where is the image formed?
2. In a glass-slab experiment, a student measures `i = 45°`, `r = 28°` and `e = 45°`. (a) What does `i = e` tell you? (b) Why is the emergent ray not on the original straight line? (c) Give one precaution for fixing pins.
3. A lens of power `+2 D` forms a sharp image of a distant tree on a screen. (a) What kind of lens is it? (b) What is its focal length? (c) At what distance from the lens should the screen be placed?

14. Answer key and hints

14.1 MCQ answers

Table 14.1 — MCQ key

Q	Answer	One-line reason
1	b	A concave mirror can form a real enlarged image when the object is between C and F.
2	b	Along the normal, the angle with the normal is 0°.
3	a	A convex mirror has a virtual focus behind it, so `f > 0`.
4	b	Lens power is measured in dioptres (D).
5	a	Air to glass is rarer to denser; speed decreases and the ray bends towards the normal.
6	b	The lens formula is `1/v - 1/u = 1/f`.
7	c	A concave lens always gives virtual, erect and diminished images for a real object.
8	b	Negative `m` means inverted; `|m| = 2` means enlarged.
9	b	`n = c/v`; larger `n` means smaller `v`.
10	b	`f = 1/P = 1/4 = 0.25 m`, positive for a convex lens.
11	b	Two parallel faces make the emergent ray parallel but laterally shifted.
12	c	`+2 + (+1) = +3 D`, a positive net power.

14.2 Assertion–Reason answers

1. **A** — a convex mirror gives an erect small image and covers a wide field.
2. **A** — the decrease in speed causes bending towards the normal.
3. **A** — `P = 1/f` and a concave lens has `f < 0`.
4. **D** — the Assertion is false (for a real inverted image, `m < 0`); the Reason is true.
5. **A** — equal and opposite changes at the two parallel faces make `i = e`.
6. **B** — both statements are true, but the optical-centre ray rule does not explain diamond's refractive index.

14.3 Short-answer key points

1. Same plane; `i = r`; angles are measured from the normal.
2. Always erect image; small image gives a wider field of view.
3. F is where parallel rays meet; `f = PF`; `R = 2f`.
4. `sin i/sin r` is constant; more generally `n1 sin i = n2 sin r`; `n = c/v`.
5. The two faces are parallel, so the bending on entry is reversed on exit; `i = e`.
6. One dioptre is `1 m^-1`; `P = 1/f` and `f < 0` for a concave lens, so `P < 0`.
7. Real: rays actually meet, can be caught on a screen, usually inverted. Virtual: rays appear to meet, cannot be caught, usually erect.
8. A source at F gives a parallel reflected beam, concentrating light in front of the vehicle.
9. No. At normal incidence, speed and wavelength change but the direction does not change, so there is no visible bending.

14.4 Numerical answers with working

Question 1

`u = -25 cm`, `f = -15 cm`. `1/v = -1/15 + 1/25 = -2/75`, so `v = -37.5 cm`. `m = -v/u = -1.5` and `h' = -6 cm`. **Real, inverted, 6 cm tall image 37.5 cm in front.**

Question 2

$f = +20$ cm, $u = -30$ cm. $1/v = 1/20 + 1/30 = 1/12$, so $v = +12$ cm; $m = -12/(-30) = +0.4$. **Virtual, erect image 12 cm behind the mirror.**

Question 3

$v = c/n = (3 \times 10^8)/1.50 = 2 \times 10^8$ m/s.

Question 4

$n \approx \sin 60°/\sin 35° = 0.866/0.574 \approx 1.51$.

Question 5

$u = -20$ cm, $f = +15$ cm. $1/v = 1/15 - 1/20 = 1/60$, so $v = +60$ cm. $m = 60/(-20) = -3$; $h' = -9$ cm. **Real, inverted, 9 cm tall image.**

Question 6

$u = -45$ cm, $f = -15$ cm. $1/v = -1/15 - 1/45 = -4/45$, so $v = -11.25$ cm; $m = (-11.25)/(-45) = +0.25$. **Virtual, erect, quarter-sized image.**

Question 7

$P_{total} = +5 + (-2) = +3$ D; $f = 1/3$ m $\approx +0.333$ m $= +33.3$ cm. **Converging combination.**

Question 8

$f = -80$ cm $= -0.80$ m; $P = 1/f = -1.25$ D. **Concave/diverging lens.**

14.5 Diagram/case-based answers

1. (a) Between F and P. (b) A ray parallel to the axis that reflects through F, and a ray striking P that reflects with $i = r$; extend them behind the mirror. (c) Behind the mirror, virtual, erect and enlarged.
2. (a) The emergent ray is parallel to the incident ray. (b) It is laterally displaced by refraction at the two faces. (c) Keep pins vertical and separated; do not move the slab; use a sharp pencil.
3. (a) Convex/converging. (b) $f = 1/P = 0.5$ m $= 50$ cm. (c) About 50 cm from the optical centre, where a distant object forms a sharp image at F2.

15. Common mistakes and final checklist

15.1 Mistake → correction

Table 15.1 — Fix these before the exam

Common mistake	Correction
Measuring `i` and `r` from the mirror surface.	Always measure from the **normal**.
Using `1/v - 1/u = 1/f` for a mirror.	Mirror formula has a **plus**: `1/v + 1/u = 1/f`.
Using `-v/u` for lens magnification.	Lens magnification is `v/u`; mirror magnification is `-v/u`.
Taking `f = R` instead of `R/2`.	For a spherical mirror, `f = R/2`.
Giving `f` in cm inside `P = 1/f`.	Convert to **metres** first, or use `P = 100/f(cm)`.
Calling the image real just because `v` is positive.	For a lens, `v > 0` usually means real; for a mirror, a real image has `v < 0`.
Saying a convex mirror gives an enlarged image.	It gives a **small**, virtual, erect image for a real object.
Saying a concave lens can form a real image of a real object at Class 10 level.	It always gives a virtual, erect, diminished image.
Forgetting that normal incidence still changes speed.	There is no visible bending, but the speed and wavelength still change.
Drawing backward extensions as solid rays.	Use **dashed lines** for virtual extensions and virtual images.
Mixing cm and m in one calculation.	Choose one unit and write it beside every value.

15.2 You are ready when you can...

- Explain all terms in Table 11.2 without looking.
- Draw the four mirror ray rules and three lens ray rules from memory.
- Fill the concave-mirror and convex-lens image tables from ray rules, not by rote.
- Write both formulas, both magnifications and the power formula with correct signs.
- Solve a numerical in five clear steps and describe the image in the last line.
- Explain the three practical activities, their results and two precautions each.
- Correct every item in the mistake table.

Final exam habit
In a diagram question, label **P/O, F, C or 2F**, put arrows on rays, and write the image nature below the diagram. Even a correct drawing can lose marks if it is unlabelled.

15.3 Sources and note

- NCERT, **Science — Textbook for Class X**, chapter Light — Reflection and Refraction.
- CBSE, **Science (Subject Code 086), Class X Curriculum 2026–27**, Unit III: Natural Phenomena.
- NCERT/CBSE **Science Laboratory Manual** and **Exemplar Problems** for practical and higher-order practice.
- These are **original study notes** written in simple English. They explain the NCERT chapter; they are not a copy of the NCERT textbook.

Best of luck! Revise the diagrams once, practise the signs twice, and the numericals will become easy.